REPURPOSING COLLABORATION OPPORTUNITY

CINACALCET -> autosomal dominant hypocalcemia 1

For clinical-collaborator outreach.

Why this hypothesis is real

Drug-target convergence on the disease scores **0.83** (noisy-OR over OT target-disease associations). This pair is fully novel relative to all known drug-indication assignments in ChEMBL. Therapeutic direction is **aligned** with OT genetic evidence (the drug pushes the target the disease-protective way). Lead target is well expressed in the disease-relevant tissue (pancreas). Orthologous-gene model-organism evidence is strong (phylo_score 0.63, n=3).

Why now (IP runway + literature footprint)

Loss-of-exclusivity year **2026**, with 42 generic ANDAs already on the market.

Clinical opportunity

US population: approximately **1,665** patients -- below the 200,000 FDA Orphan Drug Act threshold. Orphan exclusivity + premium pricing changes the deal math materially. (source: Orphanet Annual incidence (1-9 / 1 000 000, Europe))

What we propose

BioBix (Department of Mathematical Modelling, Statistics and Bioinformatics, Ghent University) brings bioinformatics support to the collaboration:

- Target validation across blood / immune / cancer multi-omics datasets
- Biomarker candidate identification from public + private cohorts
- Patient stratification analytics for trial-enrichment design
- Mechanism-of-action mining via systems-biology priors
- Quick-turn evidence syntheses for IIT and partnered programs

Proposed collaboration model:

- Investigator-initiated trial framework with the substance owner
- BioBix as bioinformatics analytics partner (target/biomarker/biostatistics)
- Joint publication strategy; data-sharing agreement gated on IP terms

Next step

A 30-minute scoping call to align on feasibility. We would come prepared with a brief mechanistic dossier, an annotated literature pull, and a pilot biomarker shortlist for the disease.

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